Alkanes

Exam code: 0620 & 0971

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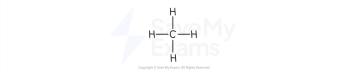
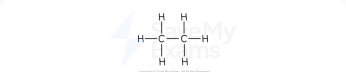
Alkanes: properties & bonding

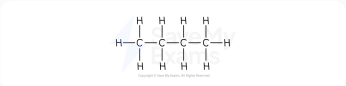
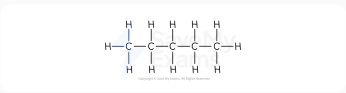
What is an alkane?

- Alkanes are a group of **saturated** hydrocarbons
- The term saturated means that they only have single carbon-carbon bonds, there are no double bonds
- Alkanes have covalent bonds as they consist of non-metal atoms
- The general formula of the alkanes is $\mathbf{C_nH_{2n+2}}$

Table of alkanes

Displayed formula	Name	Molecular formula
	methane	CH_4
	ethane	C_2H_6

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

	butane	C_4H_{10}
	pentane	C_5H_{12}

The first five members of the alkane homologous series

Examiner Tips and Tricks

Co-ordinated students only need to be able to name and draw the displayed formula for methane and ethane but extended students should know the whole table.

- Alkanes are colourless compounds which have a gradual change in their physical properties as the number of carbon atoms in the chain increases
- They are generally unreactive compounds but they:
  - Undergo **combustion**
  - Can be **cracked** into smaller molecules
  - React with **halogens** in the presence of light in substitution reactions

Combustion of alkanes

- Alkanes undergo complete combustion to form carbon dioxide and water:

$$\text{Alkane} + \text{oxygen} \rightarrow \text{carbon dioxide} + \text{water}$$

- Combustion of methane:

$$CH_4\,(g) + 2O_2\,(g) \rightarrow CO_2\,(g) + 2H_2O\,(l)$$

- Combustion of ethane:

$$C_2H_6\,(g) + 3\tfrac{1}{2}O_2\,(g) \rightarrow 2CO_2\,(g) + 3H_2O\,(l)$$

- Combustion of propane:

$$C_3H_8\,(g) + 5O_2\,(g) \rightarrow 3CO_2\,(g) + 4H_2O\,(l)$$

- Combustion of butane:

$$C_4H_{10}\,(g) + 6\tfrac{1}{2}O_2\,(g) \rightarrow 4CO_2\,(g) + 5H_2O\,(l)$$

alkanes with halogens

Extended tier only

- In a **substitution reaction**, one atom (or group of atoms) is swapped with another atom (or group of atoms)
- Alkanes undergo a substitution reaction with halogens in the presence of ultraviolet radiation (sunlight is a source of UV radiation)
- This is called a **photochemical** reaction
- The UV light provides the activation energy, E_a, for the reaction
- A hydrogen atom is replaced with the halogen atom
- More than one hydrogen atom can be substituted depending on the amount of ultraviolet radiation there is

In the presence of ultraviolet (UV) radiation, methane reacts with chlorine to form chloromethane and hydrogen chloride

Examiner Tips and Tricks

You need to be able to draw the displayed and structural formulae of the products formed when chlorine replaces one hydrogen atom (also known as monosubstitution)

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